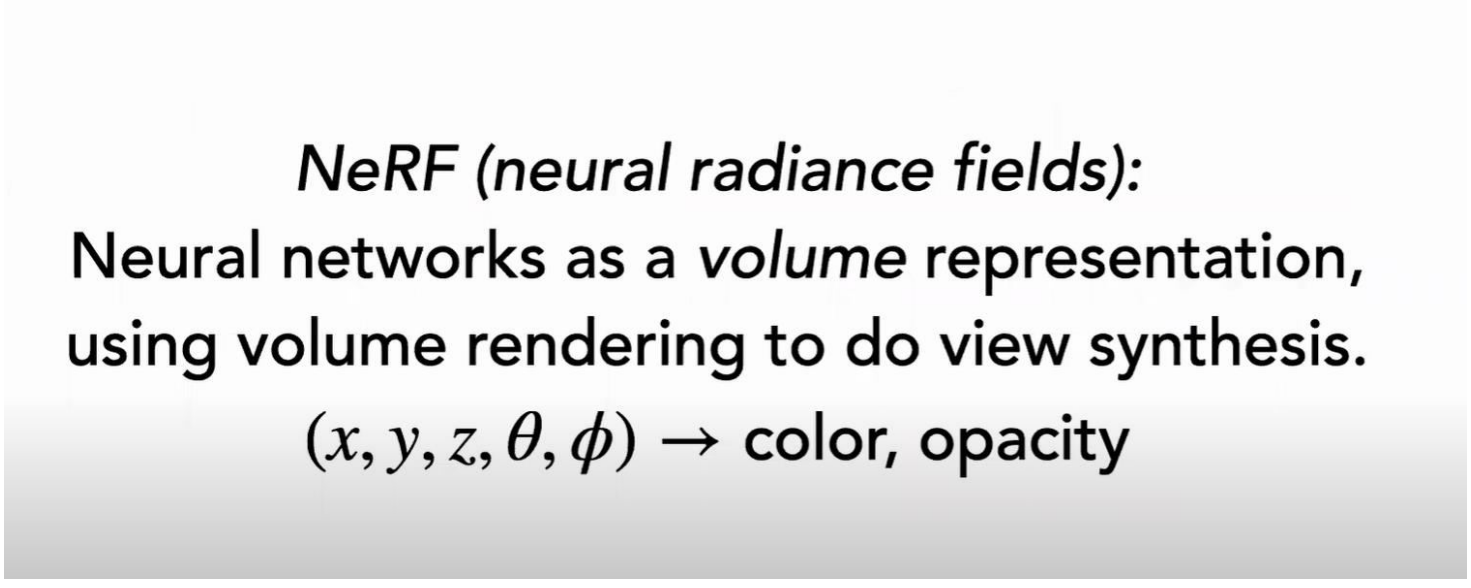
From NeRFs to Gaussians: Understanding Modern Neural Rendering

What is Neural Rendering?

- Neural rendering blends machine learning with traditional graphics.
- Generates 3D content from 2D images using learned representations.

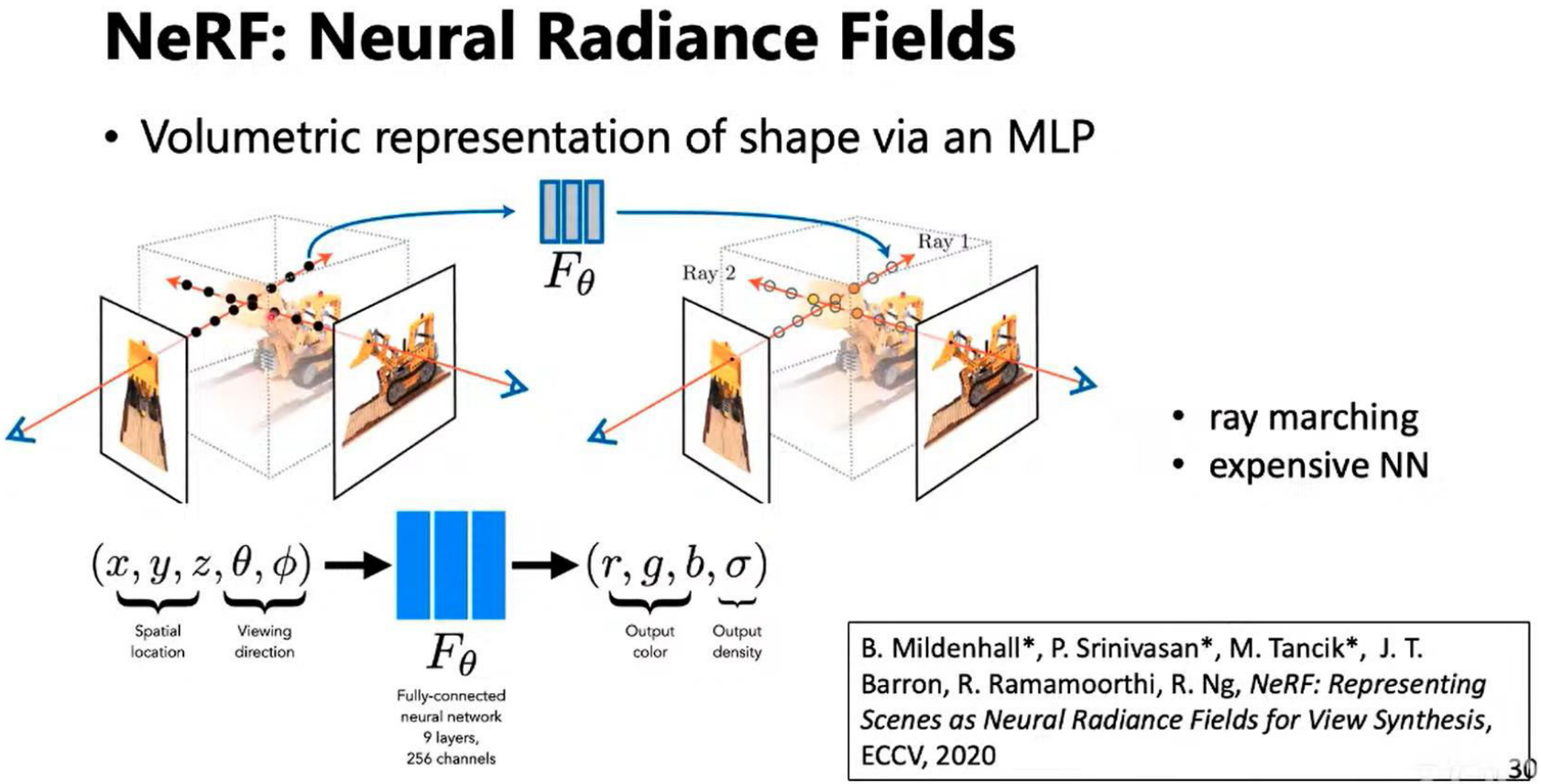
What is NeRF?

- NeRF stands for Neural Radiance Fields.
- Learns a 3D scene from 2D images by modeling a continuous volumetric field.
- Maps spatial location + viewing direction → color and density.



How NeRF Works

- Ray marching through the scene volume.
- Neural network estimates density + color at each sample.
- Composite the ray using volume rendering equation.



NeRF Strengths & Limitations

- ✅ High-quality novel view synthesis
- ❌ Slow training and rendering
- ❌ Requires dense input views

Enter Gaussian Splatting

- A recent alternative to NeRF.
- Represents scenes using 3D Gaussians with size, color, and opacity.
- Uses rasterization instead of ray marching.

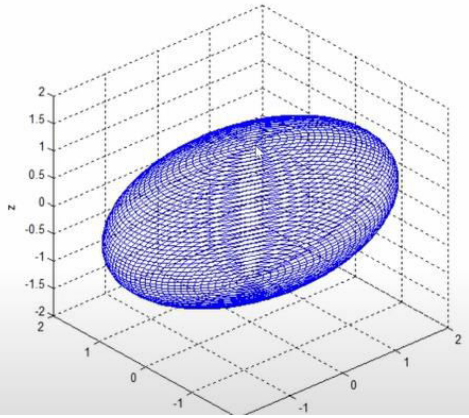
How Gaussian Splatting Works

- Each Gaussian: position, orientation, scale, color, opacity.
- Project Gaussians into image space.
- Rasterize using screen-space splatting.
- Fast & GPU-friendly.

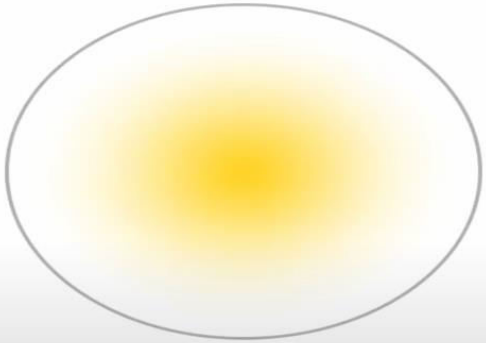


What Are Gaussian Splats?

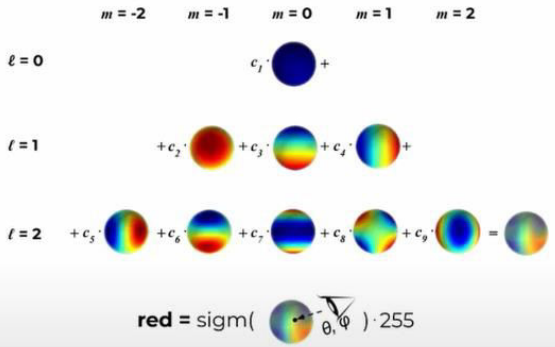
3D ellipsoids



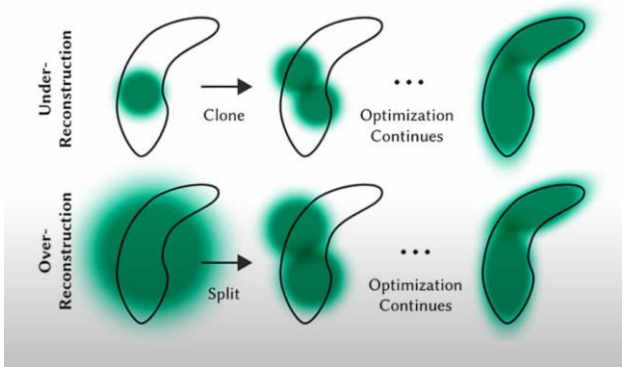
Gaussian distributed density



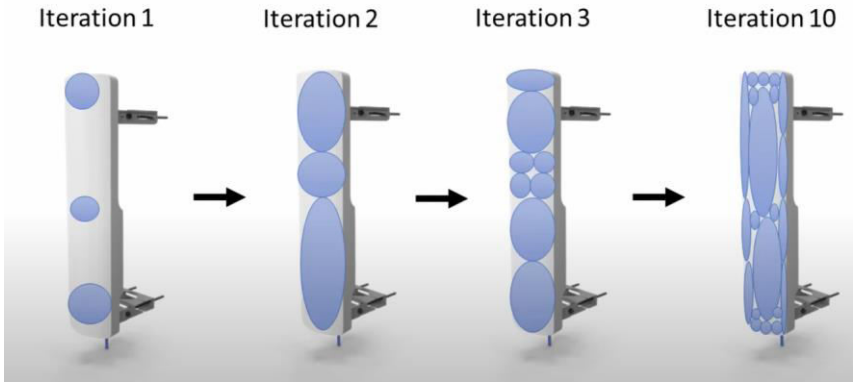
View dependent color



How are they created

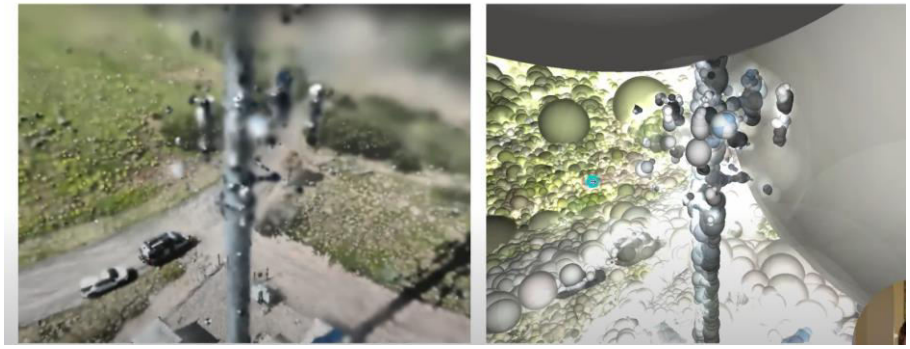


How are they created



Training

Iteration 100 – 214,127 splats



Iteration 1000 – 408,957 splats



Iteration 3000 – 2,599,683 splats



NeRF vs. Gaussian Splatting

Feature	NeRF	Gaussian Splatting
Representation	Implicit (MLP)	Explicit (Gaussians)
Rendering	Ray marching	Rasterization
Speed	Slow	Real-time
Quality	High	Very High
Training Time	Hours	Minutes

Use Cases

- 3D reconstruction
- Virtual reality
- Film & game environments
- Robotics and simulation

Additional Resources

- https://www.youtube.com/watch?v=DjOqkVIlEGY
- https://medium.com/towards-data-science/a-python-engineers-introduction-to-3d-gaussian-splatting-part-1-e133b0449fc6
- https://www.youtube.com/watch?v=9p_i9nKZA4Q&list=WL&index=105